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Long-finned pilot whale (*Globicephala melas*, Traill 1809) subspecies in the Atlantic Ocean: Are there differences in their skulls?

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The head is the window for contact, perception, and communication with the world. The skull provides the framework, the organizing hub, for the head. Shape and specific characteristics of the skull are fundamentally important to understand the evolutionary biology of a species and its feeding strategies (Rommel *et al.* 2009). In delphinid species skull-shape evolution may have been indirectly influenced by several evolutionary changes (Amaral *et al.* 2009). The two currently recognized species of the genus *Globicephala* provide an interesting example of a recent and complex evolutionary history (Oremus *et al.* 2009). The long-finned pilot whale (*Globicephala melas*, Traill 1809) and the short-finned pilot whale (*Globicephala macrorhynchus*, Gray 1846) are large delphinid species, widely distributed, that can be found in both near-shore and pelagic environments (Olson 2009). Size, shape, and color

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pattern distinctions between the two species are too variable to consider them as reliable characters for discrimination. Morphological differences between the two species based on flipper length are clear: on average pectoral fins of *G. melas* are one-fifth the body length, whereas on *G. macrorhynchus* they are one-sixth the body length (Olson 2009). Efforts for identifying free-ranging pilot whales in the northwest Atlantic region have been made using a simple photograph-based approach, based on relative morphological features and pigmentation patterns (Rone and Pace 2012). Overall, apart from pectoral fin length, the only distinctive aspect between the species is reflected in the skull (van Bree 1971). On one hand, the long-finned pilot whale has a narrower skull, with the premaxillae leaving uncovered 1 cm of the lateral borders of the maxillae, and 9–12 teeth in each row. On the other hand, the short-finned pilot whale has a shorter and broader skull, with the premaxillae covering the maxillae, and 7–9 teeth in each row (Olson 2009). Phylogenetic analyses of mitochondrial DNA (mtDNA) also support the species status (May-Collado and Agnarsson 2006, Oremus *et al.* 2009).

Long-finned pilot whales inhabit the cold temperate waters of the North Atlantic and the Southern Hemisphere, comprising two antitropical populations that are geographically isolated (Olson 2009). The first to suggest a taxonomic difference between these populations was Rayner (1939), who described them as different species on the basis of distinct color patterns. In spite of this, Davies (1960) reduced the distinction to the subspecies level: *Globicephala melas melas* in the North Atlantic and *Globicephala melas edwardii* in the Southern Hemisphere. This author argued that the last contact between the two subspecies could have occurred only 10,000–15,000 yr ago during the deglaciation interval following the Last Glacial Maximum, when equatorial waters were substantially cooler. Nowadays, in the North Atlantic, the range of *Globicephala melas melas* includes the outer continental shelf of the northeastern United States north of Cape Hatteras, the waters of Greenland, Iceland, and the Barents Sea south to the Arctic Circle, as well as in the western Mediterranean Sea. On the contrary, *Globicephala melas edwardii* is circumglobal in the Southern Hemisphere, with records as far north as 14°S in the Pacific and further south than the Antarctic Convergence, with reports off the Argentinean and Chilean coasts (Clarke and Goodall 1994, Bernard and Reilly 1999, Mansilla *et al.* 2012). Oremus *et al.* (2009) also suggested a subspecies ranking for the two populations based on the disjunctive distribution and the strong mtDNA differentiation (despite the sharing of haplotypes) they found. However, these authors argued that the two subspecies would not qualify as evolutionary significant units (ESUs).

Several studies of cranial morphometry have confirmed the presence of dolphin ecotypes (*i.e.*, populations), coastal and offshore ecotypes or subspecies in many parts of the world (Jefferson and Van Waerebeek 2004, Kemper 2004, Murphy *et al.* 2006, Perrin *et al.* 2011, Costa *et al.* 2016, Wickert *et al.* 2016). In general, pilot whale investigations have focused on life history, social structure, diet, and behavioral aspects of the species (Donovan *et al.* 1993, Wenzel *et al.* 2009). Research on body and skull morphometry of the long-finned pilot whale is scarce and

limited to the North Atlantic Ocean (e.g., Davies 1960, Sergeant 1962, Bloch and Lastein 1993). The few studies of the Southern Hemisphere subspecies describe isolated stranding or sighting events (Crespo *et al.* 1985; Goodall and Macnie 1998*a, b*). Previous to this study, only three skulls of the South Atlantic subspecies were measured and published (Casinos 1981). To our knowledge, no comparison in skull morphometry between the long-finned pilot whale subspecies has been carried out.

The aim of this study was to characterize in detail the skull morphometry of the two subspecies of long-finned pilot whale (*G. melas melas* and *G. melas edwardii*) in the Atlantic Ocean and compare them in order to provide new information about their cranial structure, as well as to contribute to the taxonomic resolution of the subspecies. We hypothesized morphometric differences in the skull of both subspecies, based on the fact that they are completely isolated and on the presence of analogous skull distinction in the short- and long-finned pilot whale species (van Bree 1971), and in the eastern and western North Atlantic populations of the long-finned pilot whale (Bloch and Lastein 1993).

Morphometric data were collected from long-finned pilot whale skulls from the two subspecies, *G. melas edwardii* and *G. melas melas*. Specimens belong to the RNP collection held at the Museo Acatushún de Aves y Mamíferos Marinos (Ushuaia, Tierra del Fuego, Argentina), to the Museo del Fin del Mundo (Ushuaia, Tierra del Fuego, Argentina), and the U.S. National Museum of Natural History, Smithsonian Institution (Washington, DC).

Only adult specimens were included in this study. Three different criteria were employed to determine which of the skulls to include in the analysis: age estimation (growth layer groups technique in teeth), total length, and premaxillae-maxillae fusion. Regarding age criterion, specimens were considered as adults when they were older than 8 yr in females and 12 yr in males (Olson 2009). When total length was employed, specimens longer than 400 cm in females and 446 cm in males were considered adults (Sergeant 1962). When data on age estimation or total length were not available, specimens were judged cranially mature if distal fusion had occurred between the premaxillae and maxillae in the rostrum; this provides a rough assessment of maturity based on homology with small delphinids (Perrin and Heyning 1993). The sample totaled 93 adult specimens, 69 from the southern South Atlantic Ocean (*Globicephala melas edwardii* subspecies) and 24 from the North Atlantic (*Globicephala melas melas* subspecies). An exhaustive list of specimens used in the study is available as electronic supplementary material in a public repository at DOI 10.6084/m9.figshare.4828579.v1. A map of the stranding location of the individuals analyzed is shown in Figure 1.

Because most of the specimens were beach-cast or museum acquisitions (many of them were collected long ago and not as part of a dedicated research program), sex and total body length were unknown for many of the specimens; therefore, sexual dimorphism could not be studied. In this regard, as observed in sperm whales (*Physeter macrocephalus*) and orcas (*Orcinus orca*), pilot whales exhibit differences in size according to sex, where adult males are longer than females and

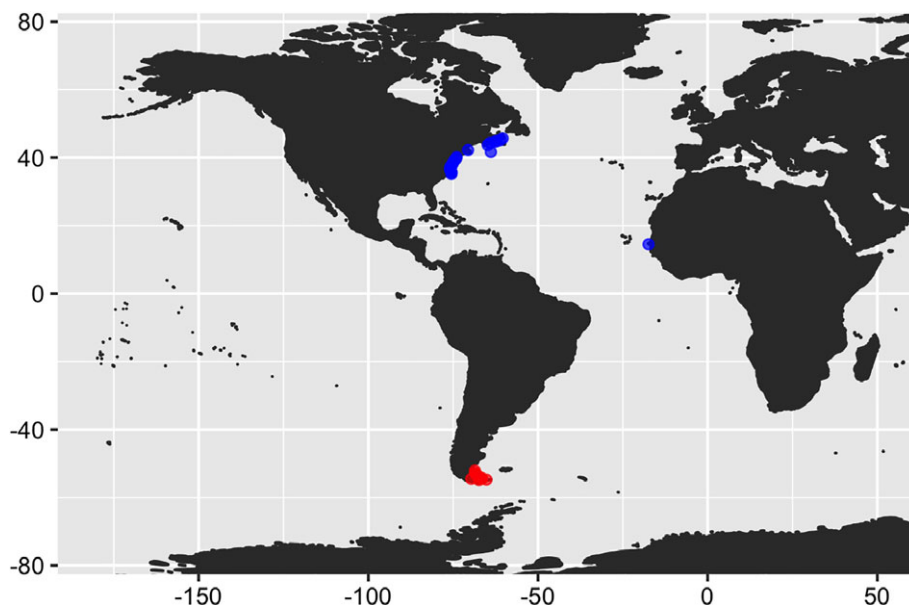


Figure 1. Map showing the stranded locations of the individuals of *G. melas edwardii* and *G. melas melas* analyzed in this study.

develop a more pronounced melon (Olson 2009). However, sexual dimorphism has not been described at the level of skull morphometrics in any *Globicephala* species, subspecies, or population. We were not able to test sexual dimorphism in the present work since data on sex was scarcely available (*i.e.*, less than 10 sexed specimens per subspecies). Therefore, skull variation is likely to have been uncontrolled with respect to the sex factor in this study. More sexed specimens should be gathered to consider sex as a factor of variance in skull morphometrics of *Globicephala melas* subspecies.

On each skull, 21 measurements were taken with an anthropometer to the nearest millimeter by TIM following Perrin (1975), with addition of three measurements: width of rostrum at 1/4 length (WRQ), length of rostrum to posterior margin of external and internal nares (REN and RIN, respectively) (Table 1, Fig. 2).

In order to assess the differences in skull metrics between subspecies, a nonparametric test (Mann-Whitney *U* test, $P < 0.05$) was conducted among all measurements, since 17 of 21 variables (measurements) did not pass the normality test (Shapiro-Wilk test; data not shown). A principal component analysis (PCA) was performed to determine positions of the specimens along major multivariate axes and to identify the variables most influential in separating any apparent clusters corresponding to the two subspecies. All characters were used. Various rotating solutions were attempted (unrotated, varimax, and quartimax), and the solution

Table 1. List of skull measurements taken on long-finned pilot whale skulls (see Fig. 2)

Character	Abbreviation	Number
Condylobasal length	CBL	1
Length of rostrum	LRO	2
Width of rostrum at base	WRB	3
Width of rostrum at 1/4 length ^a	WRQ	4
Width of rostrum at 1/2 length	WRH	5
Width of rostrum at 3/4 length	WRT	6
Width of premaxillae at 1/2 length	WPH	7
Greatest width across premaxillae	GWPM	8
Rostrum to external nares ^a	REN	9
Rostrum to internal nares ^a	RIN	10
Preorbital width	GPRW	11
Postorbital width	GPOW	12
Zygomatic width	GWZP	13
Parietal width	PARW	14
Width of external nares	GWEN	15
Width of internal nares	GWIN	16
Height of braincase	BRNH	17
Height of temporal fossa	LTF	18
Length of temporal fossa	HTF	19
Length of orbit	LOR	20
Length of antorbital process	LAPL	21

^aThese measurements were not taken by Perrin (1975).

that explained most of the variance was used. The subspecies groups in the PCA were highlighted with 95% confidence ellipses.

A discriminant function analysis (DFA) was also carried out to verify the characters that best explained the separation of the subspecies according to the PCA. The statistical significance of DFA clustering was assessed through Wilks' lambda (Quinn and Keough 2002).

In order to gain insight into morphometric implications in functional aspects of the skull and to avoid size-biased misinterpretations we standardized to the condylobasal length (CBL) and to the zygomatic width (GWZP) some of the measurements regarding length (*e.g.*, length of rostrum, LRO, rostrum to external nares, REN, among others) and width (*e.g.*, width of rostrum at base and at 1/4 length, WRB and WRQ, among others), respectively. We performed a nonparametric test (Mann-Whitney *U* test, $P < 0.05$) throughout the standardized measurements with the aim of analyzing possible differences between subspecies. All statistical analyses were done using R software (R Core Team 2017).

Specimens of the two subspecies of long-finned pilot whale (*Globicephala melas*) showed differences in several skull measurements. *G. m. edwardii* have significantly longer overall skull and rostrum, broader internal and external nares, larger (length and height) temporal fossa, and longer orbit and antorbital process than *G. m. melas* specimens. Furthermore, on average *G. melas edwardii* specimens

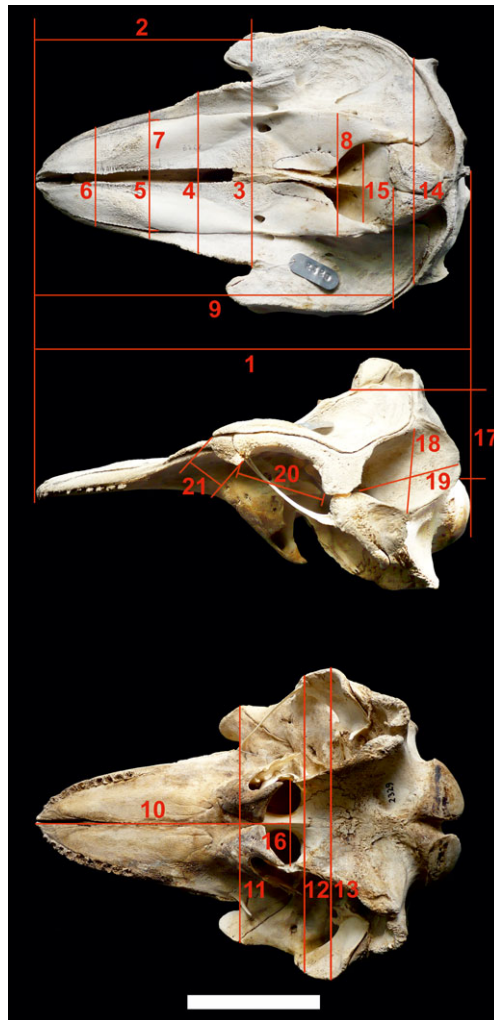


Figure 2. Dorsal, left lateral and ventral views of an adult skull of southern long-finned pilot whale, *G. melas edwardii* (RNP2329). Red lines represent the 21 measurements taken on each skull (see Table 1). Scale bar of 20 cm.

showed overall wider rostrum, wider skull across the zygomatic, premaxillae, and parietal sections (Table 2). Moreover, several standardized measurements showed significant differences between subspecies (Table 3).

With the aim of studying the length/breadth ratio of the skull, we divided the condylobasal length (CBL) by the zygomatic width (GWZP). A significant difference was found for this ratio between subspecies (Mann-Whitney U test, $P < 0.0001$). Although most measurements regarding width are similar in both subspecies (Table 2), the difference

Table 2. Summary of skull measurements (in mm) for adult specimens of subspecies *G. melas edwardii* and *G. melas melas*

Character	<i>G. melas edwardii</i>			<i>G. melas melas</i>			Mann-Whitney ^a
	Mean $\pm$ SD	Range		Mean $\pm$ SD	Range		
Condylobasal length	624.0 $\pm$ 37.9	569–711		596.0 $\pm$ 29.1	550–651		**
Length of rostrum	322.1 $\pm$ 26.8	258–393		302.4 $\pm$ 17.4	281–337		***
Width of rostrum base	240.3 $\pm$ 21.7	208–294		233.7 $\pm$ 23.0	204–297		ns
Width of rostrum 1/4 length	133.5 $\pm$ 17.9	103–174		130.2 $\pm$ 15.7	107–169		ns
Width of rostrum 1/2 length	179.0 $\pm$ 22.5	143–226		178.1 $\pm$ 21.1	152–234		ns
Width of rostrum 3/4 length	218.8 $\pm$ 25.5	180–273		217.0 $\pm$ 23.1	191–280		ns
Width of premaxillae 1/2 length	165.3 $\pm$ 20.9	136–232		155.3 $\pm$ 16.5	133–202		0
Rostrum to external nares	419.3 $\pm$ 33.9	341–493		384.5 $\pm$ 53.1	196–450		**
Rostrum to internal nares	361.8 $\pm$ 24.9	318–432		326.1 $\pm$ 19.6	299–363		***
Preorbital width	389.7 $\pm$ 37.4	339–466		392.4 $\pm$ 36.2	314–480		ns
Postorbital width	426.5 $\pm$ 37.9	376–508		421.3 $\pm$ 60.2	196–513		ns
Zygomatic width	409.0 $\pm$ 33.0	355–483		410.4 $\pm$ 32.1	365–492		ns
Width of external nares	94.1 $\pm$ 6.1	77–111		90.0 $\pm$ 7.9	79–104		0
Width of internal nares	119.7 $\pm$ 11.6	97–149		108.8 $\pm$ 9.4	96–136		***
Width across premaxillae	160.3 $\pm$ 13.0	137–190		155.7 $\pm$ 14.8	129–185		ns
Parietal width	287.3 $\pm$ 22.1	248–338		287.0 $\pm$ 20.7	246–323		ns
Height of braincase	249.5 $\pm$ 15.4	210–289		239.4 $\pm$ 14.2	204–271		**
Length of temporal fossa	146.5 $\pm$ 17.2	121–240		136.6 $\pm$ 12.7	118–169		**
Height of temporal fossa	115.6 $\pm$ 10.3	93–142		101.0 $\pm$ 19.3	61–138		***
Length of orbit	107.4 $\pm$ 7.9	88–123		97.6 $\pm$ 6.7	87–108		***
Length of antorbital process	62.4 $\pm$ 9.4	44–87		57.9 $\pm$ 7.0	43–79		0

^aNonparametric test (Mann-Whitney *U* test) between subspecies, level of significance: * < 0.05, ** < 0.01, *** < 0.001, ns not significant (*P* > 0.05).

Table 3. Skull standardized measurements to the condylobasal length (CBL) and zygomatic width (GWZP) for cranially-mature specimens

Character	Mann-Whitney ^a
Standardized to CBL	
Length of rostrum (LRO)	0
Rostrum to external nares (REN)	ns
Rostrum to internal nares (RIN)	***
Length of temporal fossa (LTF)	ns
Height of temporal fossa (HTF)	**
Standardized to GWZP	
Width of rostrum at base (WRB)	***
Width of rostrum at 1/4 length (WRQ)	ns
Width of rostrum at 1/2 length (WRT)	ns
Width of rostrum at 3/4 length (WRH)	ns
Width of external Nares (GWEN)	0
Width of internal nares (GWIN)	***

^aNonparametric test (Mann-Whitney *U* test) between subspecies, level of significance: * < 0.05, ** < 0.01, *** < 0.001, ns not significant ($P > 0.05$).

mentioned might reflect a much longer skull in the *G. melas edwardii* specimens (Fig. 3).

In the PCA (unrotated solution), the two first components accounted for 70% of the variance. Loading coefficients for these components are given in Table 4. Both subspecies overlapped almost completely on the first PC but showed separation on the second PC (Fig. 4). The characters that defined *G. melas edwardii* group (loading highest negatively) were rostrum to external and internal nares (REN and RIN), length and height of temporal fossa (LTF and HTF). Characters defining *G. melas melas* group (loading highest positively) for the second PC were width of rostrum at base (WRB) and at 1/4, 1/2, and 3/4 of its length (WRQ, WRH, WRT, respectively).

The DFA complemented the PCA results, clustering the skulls into two well-separated groups. The most important skull characters that explained the separation of the groups (WRB, WRQ, WRH, WRT, REN, RIN, LTF, and HTF) were used for the DFA; the percentage of factor scores classified to the correct group was 90%. The statistical significance of the DFA clustering was confirmed by the Wilks' lambda statistics ($\lambda = 0.025$, $P < 0.05$).

Our results clearly show that several aspects of skull morphometry differ between the two subspecies of long-finned pilot whale (*Globicephala melas*). The subspecies inhabiting the Southern Hemisphere, *G. melas edwardii*, has a longer skull, larger rostrum (*i.e.*, longer), wider nares (external and internal), larger temporal fossa and longer orbits. We failed to find a significant difference in any measure of rostrum width despite the fact that all measures were slightly greater (1–7 mm, depending on the measure) in *G. melas edwardii*. The variance was high in all measures relative to these small differences. The fact that morphometry in long-finned pilot whale subspecies skulls is different was not only

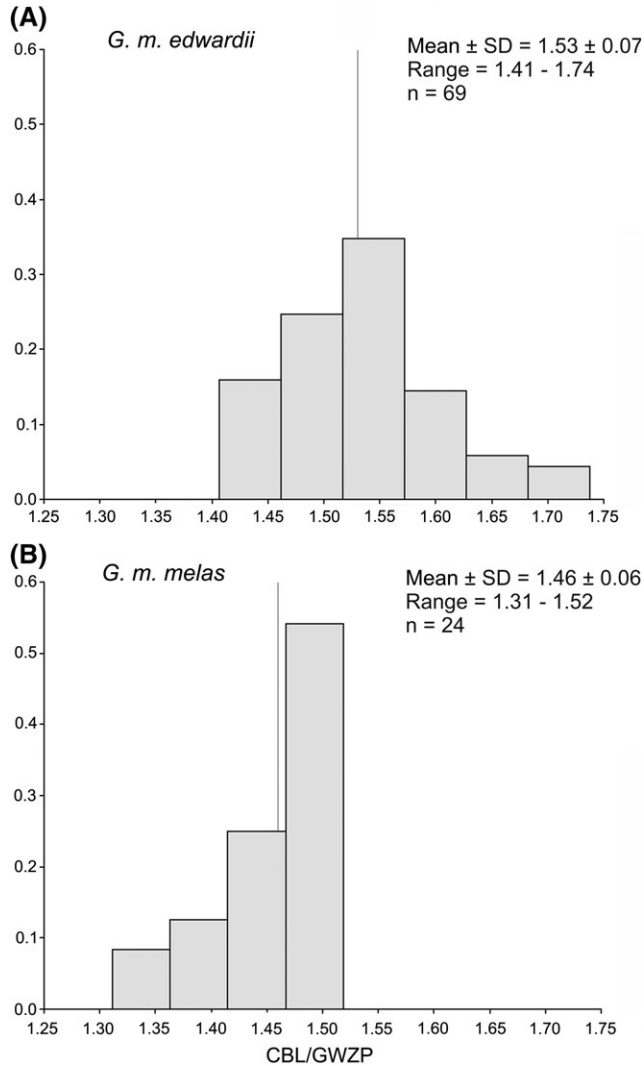


Figure 3. Histograms of condylobasal length/zygomatic width ratios (CBL/GWZP) of the two subspecies. Vertical line indicates mean.

supported by univariate analysis, but also by multivariate analyses (PCA and DFA). Accounting for 70% of the variance and 90% of correct classification, PCA and DFA results, respectively, discriminated specimens considering only a few morphometric characters of the skull: width of rostrum at base, and at 1/4, 1/2, and 3/4 length, rostrum to internal and external nares, and length and height of temporal fossa. We suggest that these characters should be taken into account in order to determine long-finned pilot whale specimens to one of the two subspecies in future skull morphometric studies.

Table 4. Factor loadings for the first two principal components. Percentage of variance accounted for each component in parentheses

Character	PC1 (64%)	PC2 (6%)
Condylobasal length	−0.94	−0.12
Length of rostrum	−0.87	−0.17
Width of rostrum at base	−0.60	0.27
Width of rostrum at 1/4 length	−0.88	0.29
Width of rostrum at 1/2 length	−0.93	0.28
Width of rostrum at 3/4 length	−0.92	0.26
Width of premaxillae at 1/2 length	−0.70	0.14
Rostrum to external nares	−0.72	−0.43
Rostrum to internal nares	−0.79	−0.37
Preorbital width	−0.93	0.25
Postorbital width	−0.87	0.19
Zygomatic width	−0.88	0.19
Width of external nares	−0.67	−0.23
Width of internal nares	−0.86	0.11
Width across premaxillae	−0.81	0.29
Parietal width	−0.76	−0.21
Height of braincase	−0.59	−0.27
Length of temporal fossa	−0.62	−0.27
Height of temporal fossa	−0.69	−0.39
Length of orbit	−0.80	0.03
Length of antorbital process	−0.83	−0.15
Variance explained	0.64	0.06

Casinos (1981) was the first to imply a difference in skull morphometry between the southwest Atlantic subspecies of long-finned pilot whale (*G. melas edwardii*) and the northern (*G. melas melas*). After having taken measurements on three skulls of pilot whales stranded on the coasts of central Argentina and southern Brazil this author suggested that the rostrum of the southwest Atlantic specimens was rather long, with the lateral borders of the maxillae not covered by the premaxillae, congruent with the conclusions of van Bree (1971). Our description of a longer skull in the Southern Hemisphere subspecies compared to its northern counterpart agrees with what Casinos (1981) reported.

Pilot whales, some of the largest odontocetes, are described as amblygnathous (Greek *amblos*, blunt, dull; *gnathos*, jaw) cetaceans with blunt head, short jaws, and reduced dentition, which suck prey directly into the oral cavity (Werth 2000). There is no doubt that not only the examination of jaws and dentition (not measured here), but also the study of skull size and shape are important for a better understanding of odontocetes' feeding habits (Werth 2000, 2006). In this work we found differences in several absolute and standardized skull characters between long-finned pilot whale subspecies and propose that they might imply distinctness in morpho-functional aspects of the cranium.

The larger temporal fossa and wider rostrum of *G. melas edwardii* suggest that it may be better adapted to feeding on larger, tougher prey

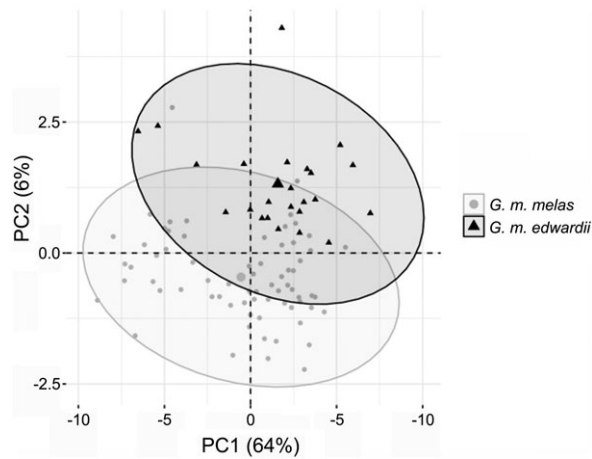


Figure 4. Plot of the PCA showing the first two principal component scores for long-finned pilot whale skulls grouped by *G. melas edwardii* and *G. melas melas* subspecies.

than *G. melas melas*. The temporal fossa serves as an attachment site for the temporal muscle and reflects the size and mass of the latter (Murie 1873). The temporal muscle is responsible for the closing of the lower jaw (Murie 1873, Perrin *et al.* 2003, Marshall 2009). This difference in the temporal fossa, indicative of a larger attachment area for the temporal muscle, may imply a stronger bite in *G. melas edwardii*. Furthermore, as force and velocity are inversely proportional (Marshall 2009), *G. m. melas* might perform greater mouth-closing velocities than *G. melas edwardii*.

Although there is a consensus that cephalopods (octopus and squid) are the main component of long-finned pilot whales diet worldwide (Gannon *et al.* 1997, Beatson *et al.* 2007, Mansilla *et al.* 2012, Santos *et al.* 2014), three dietary patterns have been suggested for different populations inhabiting North Atlantic waters (*G. m. melas*): (1) diverse diet (≥ 10 prey species) dominated by squid (Gales and Pemberton 1992, Gannon *et al.* 1997); (2) restricted diet (≤ 3 species) dominated by squid (Sergeant 1962); and (3) restricted diet (≤ 3 species) dominated by fish (Overholtz and Waring 1991). Such dietary patterns might be extended worldwide, since Beatson *et al.* (2007) proposed that *G. melas edwardii* fit the second pattern, one of low dietary diversity dominated by squid. Despite the fact that squid species represent the principal target for long-finned pilot whales, literature data (based on stomach contents and stable isotope analyses) indicate that the identity of the prey items ingested by each pilot whale subspecies is not the same. While *G. melas melas* specimens off the western North Atlantic prefer to hunt on two species of squid (*Loligo pealeii* and *Illex illecebrosus*) and a species of fish (*Scomber scombrus*) (Sergeant 1962, Overholtz and Waring 1991, Abend and Smith 1997, Gannon *et al.* 1997); *G. melas edwardii* seems to have a broader diet based on squid species: *Nototodarus* spp.

(Beatson *et al.* 2007), *Loligo gahi* and *Moroteuthis ingens* (Clarke and Goodall 1994, Mansilla *et al.* 2012), *Lycoteuthis lorigera* and *Histioteuthis* spp. (Dos Santos and Haimovici 2001). In South America, *G. melas edwardii* diet is comprised exclusively of squid (Clarke and Goodall 1994, Dos Santos and Haimovici 2001, Mansilla *et al.* 2012). Such a difference in the diet, evidenced by variation in prey items but not by prey size, is usually reflected in the morphology of odontocetes (Berta *et al.* 2006), and might be one of the sources for important differences in some characters of the skull between the two subspecies of long-finned pilot whale. Several investigations have suggested similar functional implications after analyzing skull characters in many odontocetes: *Tursiops truncatus* (Perrin *et al.* 2011), *Pontoporia blainvillei* (Del Castillo *et al.* 2014), *Stenella coeruleoalba* (Conry *et al.* 2016), and in species of the Phocoenidae family (Galatius *et al.* 2011). Furthermore, it has been demonstrated for other large social odontocetes, such as orcas, that adaptation to ecological variables (*e.g.*, food) may lead to discontinuous ecological or phenotypic stocks, representing isolated adapted units (Foote *et al.* 2009). We propose that feeding specializations may have occurred in long-finned pilot whale subspecies and these might explain modifications in the form and length of the skull.

Important differences found in external and internal nares width between the Southern Hemisphere subspecies *G. melas edwardii* and the North Atlantic one *G. melas melas* might relate to efficiency of air exchange in deep diving, as suggested by Perrin *et al.* (2011) between coastal and offshore ecotypes in the common bottlenose dolphin (*Tursiops truncatus*). Considering that the nasal and pterygoid air sacs, whose function is to regulate respiratory gases and to protect cavities and cartilages from pressure changes during diving, are associated with the external and internal nares, respectively (Reidenberg and Laitman 2008), wider bony nares observed in *G. melas edwardii* might imply greater airflow and gas exchange. This may lead to better diving capabilities in the Southern Hemisphere subspecies. Analogously, the suggested relation between nare size and diving capability emerges if long- and short-finned pilot whale species are compared. In this regard, long-finned pilot whales inhabit deep waters as they mainly feed on prey items that are found at great depths (Desportes and Mouritsen 1993, Baird *et al.* 2002). However, dives deeper than 828 m and longer than 28 min rarely occur in this species (Baird *et al.* 2002, Heide-Jørgensen *et al.* 2002, Nawojchik *et al.* 2003, Mate *et al.* 2005). By contrast, short-finned pilot whales (*Globicephala macrorhynchus*) perform much deeper and longer dives than their long-finned relatives, with maximum depths of 1,500 m and dive duration exceeding 40 min (Wells *et al.* 2013). Short-finned pilot whale skulls are larger than long-finned pilot whale skulls (van Bree 1971, Olson 2009) and, as a consequence present wider internal nares being a possible explanation for higher dive capabilities (deeper maximum depths and longer dives). Studying the North Atlantic subspecies (*G. melas melas*), Heide-Jørgensen *et al.* (2002) equipped individuals with satellite-linked time-depth recorders and suggested that pilot whales either

have lower dive capabilities than other similarly sized odontocetes or, more likely, that they use a niche in the water column that requires less extreme diving in terms of dive duration and maximum dive depth.

There is a lack of information concerning diving behavior in long-finned pilot whales of the Southern Hemisphere. A direct comparison between long-finned pilot whale subspecies diving patterns (*e.g.*, dive duration and depth, descent/ascent speeds, among others), might add support to our hypothesis of dissimilar diving capabilities due to differences in nares width.

A potential functional implication for the significant difference in the width of the external and internal nares and the premaxillae found in long-finned pilot whale subspecies might be related to sound production. The morphology of the soft tissue responsible for sound production is conditioned, in part, by the structure of the nares and premaxillary bones (Evans and Prescott 1962, Crandford *et al.* 1996). Wider nares and premaxillae observed in *G. melas edwardii* specimens could thus be associated with better vocalization capabilities, as has been proposed for other delphinids (Perrin *et al.* 2011, Del Castillo *et al.* 2014, Conry *et al.* 2016). The air sacs connected to the nasal passages (*i.e.*, external and internal nares) are important in shaping the directionality of outgoing biosonar pulses (Amundin and Andersen 1983, Aroyan *et al.* 1992, Crandford *et al.* 1996). Wider nares and premaxillae imply larger air sacs (*i.e.*, nasofrontal and premaxillary sacs), and given the nature of odontocetes to regulate the amount of air in the air sacs (Moore *et al.* 2008, Au *et al.* 2010), this suggests a greater capacity in sound production and a better accuracy in directing the pulses.

Pilot whale feeding may sometimes require a high level of coordination between group members, behavior that is achieved by the production of pulsed sounds and complex whistles. Feeding events are characterized by fast, very intense, and continuous clicking (Weilgart and Whitehead 1990). In odontocetes, vocal output has been shown to vary with feeding strategy (Barret-Lennard *et al.* 1996, Madsen *et al.* 2004, Gridley *et al.* 2015). This supports the hypothesis of distinct and probably better echolocation and sound production abilities in *G. melas edwardii*, considering that this subspecies has greater diet diversity that may require different sound strategies (*i.e.*, production of tonal calls, clicks, burst pulses) for succeeding in prey capture.

This is the first study that characterizes and compares the skull morphometry of the two long-finned pilot whale subspecies in detail. Our results lead to the conclusion that these subspecies exhibit significant differences in skull morphology in the Atlantic Ocean. We have suggested functional implications for feeding habits, diving behavior and sound production that should be tested in future studies. Since our analysis was restricted to the Atlantic Ocean, an interesting forthcoming step would be to extend the present study to other populations of the Southern Hemisphere. More comparative research like the present work should be considered in order to help reassess the taxonomy of long-finned pilot whales and to improve our understanding of the

evolutionary biology of this species, thus providing useful tools for future conservation issues.

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